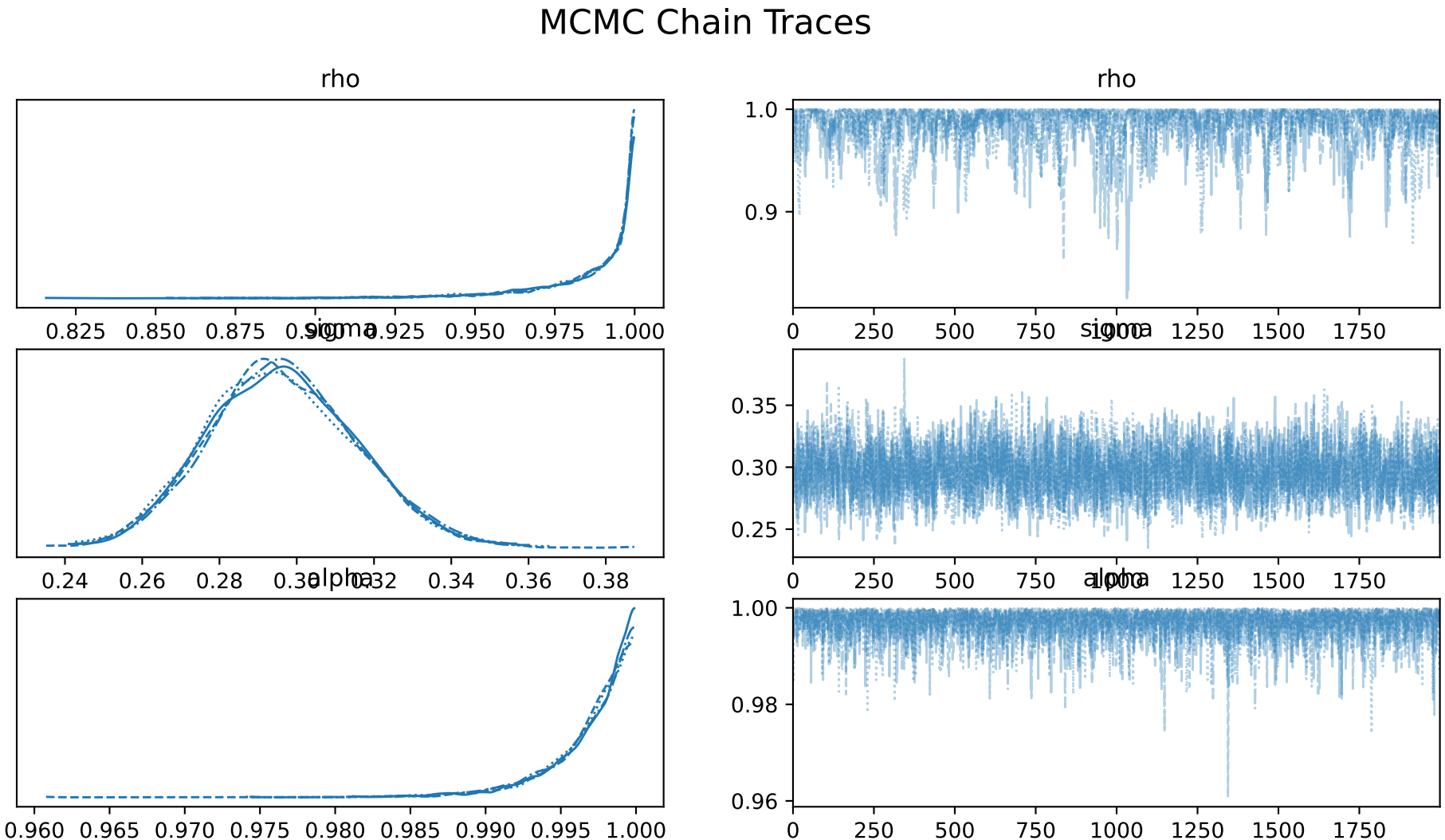
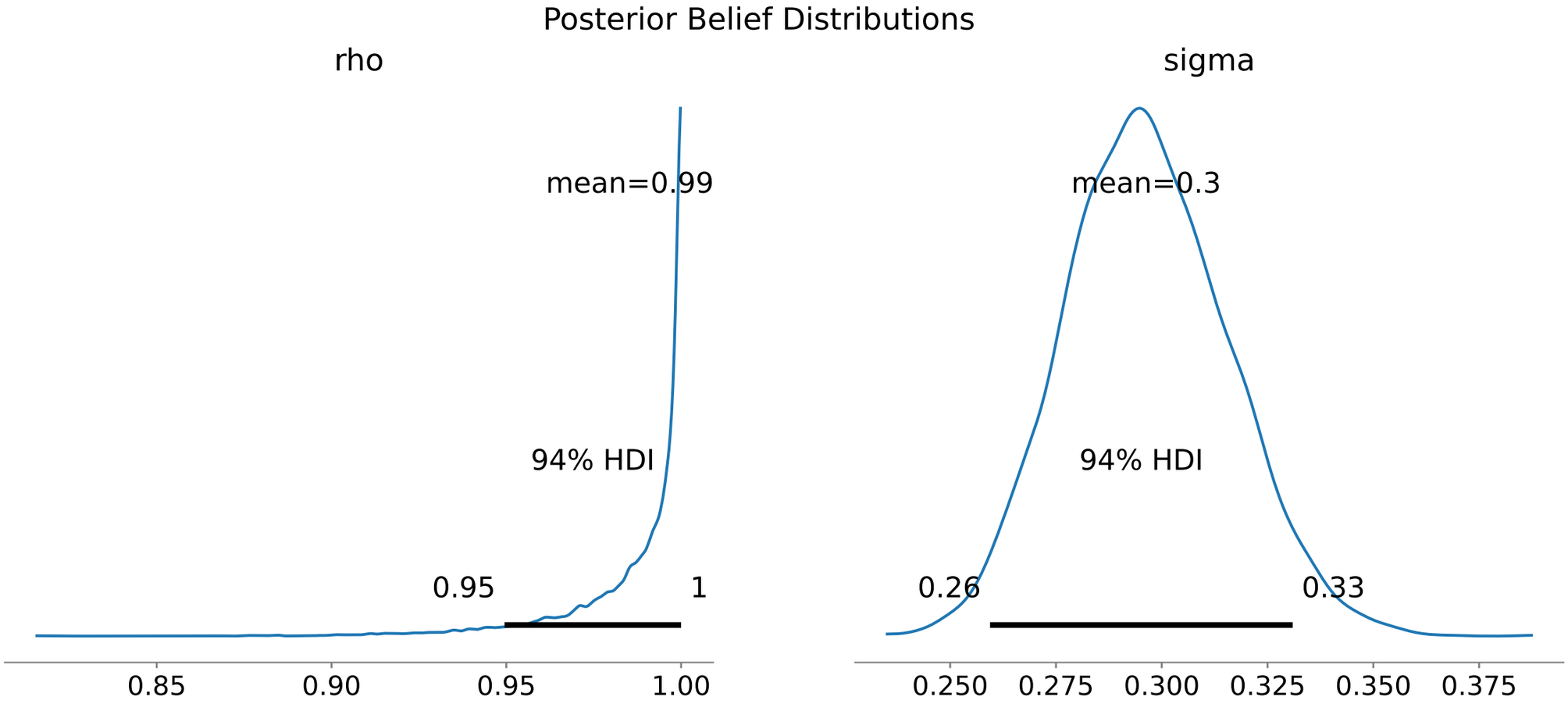
Bayesian CAR Model Summary

```
======================================================================
CAR MODEL EXECUTION REPORT | idata_car_population_v3.nc
Structure: 4 chains | 1500 warmup | 2000 samples
======================================================================
PARAMETER    |     MEAN |    R-HAT |   ESS_BULK
----------------------------------------------------------------------
rho          |    0.985 |    1.010 |        593
sigma        |    0.296 |    1.000 |       2280
alpha        |    0.997 |    1.000 |       2882
b0           |   -7.708 |    1.000 |       9458
----------------------------------------------------------------------
STABILITY CHECK: PASS
MAX R-HAT:       1.010
DIVERGENCES:     0
SPATIAL FRACTION (ρ): 98.50%
======================================================================
```


MCMC Chain Traces
rho
0.825 0.850 0.875 0.900 0.925 0.950 0.975 1.000
rho
1.0
0.9
0 250 500 750 1000 1250 1500 1750
sigma
0.24 0.26 0.28 0.30 0.32 0.34 0.36 0.38
sigma
0.35
0.30
0.25
0 250 500 750 1000 1250 1500 1750
alpha
0.960 0.965 0.970 0.975 0.980 0.985 0.990 0.995 1.000
alpha
1.00
0.98
0.96
0 250 500 750 1000 1250 1500 1750


Posterior Belief Distributions
rho
mean=0.99
94% HDI
0.95
1
0.85
0.90
0.95
1.00
sigma
mean=0.3
94% HDI
0.26
0.33
0.250
0.275
0.300
0.325
0.350
0.375